

Project Proposal Draft: Audio-Qformer - A Novel Architecture for Bridging Audio and Text Modalities

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Abstract

The domain of audio processing faces a significant challenge in bridging the gap between high-dimensional, continuous audio signals and the discrete, semantic-rich representations required by modern Large Language Models (LLMs). This project proposes the development of "Audio-Qformer," a novel architecture adapted from the Q-Former successfully used in vision-language models. The core idea is to utilize a set of learnable query vectors to distill salient semantic information from raw audio waveforms into a fixed-length sequence of vectors. This distilled representation will serve as a versatile interface between a frozen audio encoder and a frozen LLM, enabling advanced downstream tasks such as zero-shot audio classification, audio captioning, and audio-text retrieval. This proposal outlines the project's objectives, a review of foundational literature, and the proposed methodology for implementation and evaluation.

1 Introduction

Recent advancements in artificial intelligence have been dominated by the success of Large Language Models (LLMs), which exhibit remarkable capabilities in understanding and generating human language. A parallel trend has emerged in multi-modal learning, seeking to connect these powerful text-based models with other modalities like vision and audio. While vision-language models (e.g., BLIP-2, Flamingo) have made significant strides, the audio domain remains comparatively underexplored.

Raw audio is fundamentally different from text; it is a continuous waveform, dense with information, and lacks the clear segmentation of words or tokens. To effectively leverage LLMs for audio tasks, a "translator" module is required. This module must be capable of processing a long, variable-length audio stream and "distilling" its most important semantic components into a compact, fixed-length representation that an LLM can understand.

This project is based on the "Audio-Qformer" concept, which adapts the Q-Former module—originally designed for images—to the audio domain. The Q-Former acts as an information bottleneck, using a small set of learnable queries to interact with features from a powerful, pre-trained audio encoder. The output of these queries forms a concise summary of the audio, which can then be fed into an LLM. Our work will focus on implementing this architecture and evaluating its efficacy in bridging the gap between continuous audio signals and discrete textual representations.

2 Objectives and Expected Outcomes

This project aims to design, implement, and validate the Audio-Qformer architecture. The specific objectives and their measurable outcomes are detailed below.

2.1 Objectives

1. **Design and Implement Audio-Qformer:** To adapt the Q-Former architecture from the vision domain to effectively process outputs from pre-trained audio encoders (e.g., BEATs, HTS-AT). This involves designing the cross-attention mechanism between learnable queries and audio features.
2. **Integrate with LLMs:** To develop a training framework that bridges a frozen audio encoder with a frozen LLM using the Audio-Qformer as the sole trainable intermediary.
3. **Train the Model:** To train the Audio-Qformer on large-scale paired audio-text datasets (e.g., AudioCaps, CLOTHO) using a two-stage strategy: 1) audio-text representation learning and 2) generative pre-training to align the Q-Former’s output with the LLM.
4. **Evaluate Performance:** To quantitatively and qualitatively evaluate the model’s ability to perform zero-shot audio understanding tasks, specifically focusing on audio captioning and audio classification.

2.2 Expected Outcomes (Deliverables)

1. **Open-Source Codebase:** A complete PyTorch implementation of the Audio-Qformer model, including training and evaluation scripts, will be made available in a public GitHub repository.
2. **Pre-trained Model Weights:** The trained weights of the final Audio-Qformer model, compatible with standard pre-trained audio encoders and LLMs, will be released.
3. **Benchmark Results:** A comprehensive report (e.g., a "results.md" file) detailing the model’s performance on standard benchmarks (e.g., AudioCaps, ESC-50, VGGSound) and comparing it against existing baselines.
4. **Final Project Report:** A final, detailed report (minimum 10 pages) documenting the architecture, training methodology, experimental setup, results, and analysis, which will serve as the final submission for this project.

3 Related Literature

The proposed work is built upon foundational research in multi-modal learning, particularly in vision-language models and recent advancements in audio-language modeling.

- **Li, J., et al. (2023). "BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models."** [1] This is the foundational paper for our project. It introduces the Q-Former as a lightweight and highly effective module to bridge a frozen image encoder with a frozen LLM. It demonstrates a two-stage pre-training strategy that we plan to adapt for the audio domain. Our work will be a direct application and adaptation of this core concept.
- **Li, L., et al. (2023). "Q-Former: A Bidirectional Transformer for Visual and Language Understanding."** [2] This work provides a deeper exploration of a Q-Former architecture (distinct from BLIP-2’s) as a bidirectional transformer, further validating the power of query-based mechanisms for modality bridging. It provides alternative design insights for our own Q-Former implementation.
- **Dong, Q., et al. (2024). "Audio-GPT: Understanding and Generating Speech, Music, Sound, and Talking Head."** [3] Audio-GPT represents a state-of-the-art, large-scale model for unified audio understanding and generation. It serves as a high-level benchmark for

what a comprehensive audio-language model can achieve. Our Audio-Qformer aims to provide a more modular, lightweight, and efficient alternative for understanding tasks, rather than a monolithic generative model.

- **Khurana, S., et al. (2024). "Speech-LLaMA: A Large Language Model for Speech."** [4] This paper explores the direct connection of speech encoders to LLMs. Its focus is primarily on speech-based tasks. Our project differentiates itself by aiming for a *general* audio model, capable of processing not just speech but also music, environmental sounds, and complex acoustic scenes.
- **Chang, H., et al. (2024). "SALMON: Self-Alignment with Instruction-Following and Multi-Modal Grounding for Speech Large Language Models."** [5] Similar to Speech-LLaMA, SALMON focuses on speech-LLM alignment. Its contribution to instruction-following and grounding is relevant, but its scope is limited to speech. Our Audio-Qformer will instead distill *all* audio, providing a more generalized "summary" for the LLM.
- **Lee, J., et al. (2025). "Listen, Understand, and Translate: A Multitask Audio-to-Text Translation Model."** [6] This work exemplifies a strong multitask model for audio-to-text applications, specifically translation. It serves as a valuable benchmark and reference for designing the evaluation protocol for our model, particularly for audio captioning and related audio-to-text generation tasks.

References

- [1] Li, J., Li, D., Savarese, S., & Hoi, S. (2023). *BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models*. arXiv preprint arXiv:2301.12597.
- [2] Li, L., et al. (2023). *Q-Former: A Bidirectional Transformer for Visual and Language Understanding*. arXiv preprint arXiv:2308.11596.
- [3] Dong, Q., et al. (2024). *Audio-GPT: Understanding and Generating Speech, Music, Sound, and Talking Head*. Interspeech 2024.
- [4] Khurana, S., et al. (2024). *Speech-LLaMA: A Large Language Model for Speech*. Interspeech 2024.
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- [6] Lee, J., Yoon, S., Lee, H., & Kim, J. (2025). *Listen, Understand, and Translate: A Multitask Audio-to-Text Translation Model*. Proceedings of the 22nd International Conference on Spoken Language Translation (IWSLT 2025).